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General information for participants about the examination Carl von Ossietzky University of Oldenburg

Study title: *Sonification enhanced neurofeedback training: alpha waves regulation and discrimination*

Welcome to our study on: "Sonification enhanced neurofeedback training: alpha waves regulation and discrimination"! Thank you for your interest in this study.

About the study

The purpose of this study is to explore whether listening to sounds based on real-time brain activity can help people control and understand their own brain waves. This approach is known as neurofeedback, a technique where brain activity is monitored and communicated to participants in real time. We focus on alpha waves, which are linked to relaxation and focus. You will learn more about them during the study.

This study is for scientific research using electroencephalography (EEG), is not meant to diagnose or treat any medical condition.

Study location

The experiment will take place in the Neuropsychology Lab, located on the Haarentor Campus in Building A7. The exact dates and times for the measurements will be arranged with the experimenter to accommodate your preferences.

What is EEG?

As part of this study, you will undergo an electroencephalogram (EEG). An EEG tracks and records brain wave patterns via electrodes which are either attached directly to your scalp with wires or integrated in an elastic cap.

There are no risks associated with an EEG. Since the brain's potential fields are very weak at the scalp, to ensure a high-quality reading, the spots at which an electrode is attached are cleaned using a special paste and alcohol. Electrode gel is used to improve signal quality

General information for participants

between the electrode and the scalp. The chemicals used have been clinically tested and can be easily washed off after the experiment. In rare cases, skin irritations may still occur. Occasionally, pressure points can be seen for a while at the spots where the electrodes or electrode cap was attached; in very rare cases, these spots remain visible for a few days after (e.g. redness). Please let us know if you suffer from any skin allergies or hypersensitivities.

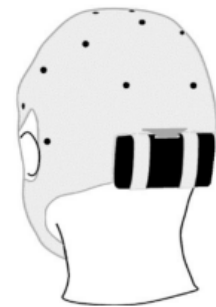
Procedure

We will meet you at the A7 entrance. Once inside the lab, you will be asked to sign the consent form and you will then complete an initial questionnaire to provide basic information, such as gender, coffee consumption or sleep quality. You will also receive a more detailed explanation of the experiment.

Before starting, you will need to wash your hair. Shampoo, towels, combs and a hairdryer will be provided.

- Mobile EEG cap

Preparing for the measurement takes about 30 minutes. At the Neuropsychology Lab, we use a mobile EEG cap, which allows for greater freedom of movement during the experiment. The compact measuring amplifier is placed at the back of the head, and the electrodes are securely connected to it.



Once the EEG cap is in place, we will begin recording your brain activity. First, there will be a baseline measurement, during three minutes, during which you will sit calmly and alternate between opening and closing your eyes based on instructions.

- Alpha waves sonification

After that, you will hear a sound that changes based on your brain activity. This is called sonification. In this study, the sound will be red noise—a deep, natural-sounding noise similar to distant cars or a waterfall. The volume of the noise reflects the strength of your alpha waves: the louder the noise, the stronger your alpha waves. If your alpha waves decrease, the noise will become quieter.

You will have time to get used to the sonification, and on the day of the experiment, you will also receive more information about alpha waves and how the sonification works

- Experiment tasks

The experiment consists of two tasks. All tasks are performed with your eyes closed, and you won't need to open your eyes at any point during the experiment. Both tasks will be repeated several times.

Before the main experiment, you will complete a short training session to help you get familiar with the tasks and practice how they work.

1. **Regulation Task:** In the first task, you will try to control your alpha waves by either increasing or decreasing them. The instructions will switch back and forth, and you will hear a bell sound when the change happens. During this task, the alpha waves sonification may be on or off.
2. **Discrimination Task:** In the second task, you will guess whether you are in a high or low alpha state. While relaxed and with your eyes closed, you will occasionally hear a bell, and then you will have to indicate your guess by pressing the upwards arrow (↑) for high or downwards arrow (↓) for low on a keyboard.

The whole experiment takes approximately 60 minutes. No painful stimuli are involved. You will be informed again of all the procedure before the recording. An experimenter will always observe and assist you during the experiment.

After the experiment, the EEG cap will be removed, and you will have the opportunity to wash your hair with shampoo and conditioner.

Recommendation

It is important that you come to the appointment well-rested. Avoid intensive exercise or alcohol the day before the measurement. If you take medication regularly, bring the patient information leaflet with you.

Voluntary participation and anonymity

Participation in the study is voluntary. You may withdraw from this study at any time without giving any reasons and without this resulting in negative consequences. Even if you withdraw from the study before the end, you are entitled to the corresponding remuneration/credit hours for test persons for the hours you participated in the study.

The data and personal notes collected as part of this study will be treated confidentially. As such, project staff members who have access to your personal data as a result of direct contact with you are bound by professional secrecy. Furthermore, the results of the study will be published anonymously, i.e. it will not be possible to identify you from the data.

Remuneration

For participating in the examination, you will receive € 12 per hour. This will be paid to you in via bank transfer, so you'll need to provide your bank account details. This information is kept completely separate from the study data.

Data protection

Categories of personal data that is processed

The following personal data is covered by data processing:

- contact details (name, address, email address, telephone numbers, etc.)
- demographic data (age, sex, etc.)
- bank account details (name, IBAN, bank, BIC, account balance, etc.)

Data processing method

Your name will only be required for the Consent Form and the Payment Form. The personal data from the questionnaires will be collected completely anonymously. Your answers and results will be recorded using a personal code word, which you create yourself based on a set of rules and that only you know. This means that no one can associate your data with your name. The anonymous data will be kept for at least 10 years. If you wish, however, you can request that all the data collected about you be deleted. For this, you will not have to tell us your name, just your code word. To create your code word, you will receive the document 'How to create your personal code word'. You keep this document. Please keep this document in a safe place as you will need it should you wish to delete your data at a later time.

Deletion periods:

Your personal data will be stored for different amounts of time. Specifically:

- Billing data: deleted 10 years after the end of the project. Reason: 10-year retention period of Division 2 acc. to Section 147 of the German Fiscal Code.
- Demographic data: made anonymous as soon as the purpose for data processing no longer applies, at the latest after 10 years (good academic practice).
- Contact details: deleted immediately after completion and evaluation of the data.

Duration of processing

After evaluation of all data and completion of the study, but at the latest once the purpose of the research ceases to exist, your data will be made anonymous as quickly as possible – in particular before publication scientific purposes takes place (e.g. specialist articles, conference papers, scientific databases [open data repositories]). The controller is obliged to do this in accordance with Section 13.2.1 of the Lower Saxony Data Protection Act (ND SG). Anonymisation means that nobody can identify you from the data. Your data is then no longer “personal” within the meaning of the legal provisions on data protection.

Use of the data

This study is for research purposes only. The personal data concerning you will not be passed on to third parties without your consent.

Contact details of the controller and data protection officer

Controller	Data protection officer
Carl von Ossietzky University Oldenburg, a public corporation, represented by the President Ammerländer Heerstr. 114-118 26129 Oldenburg Tel.: +49 441 7980 Fax: +49 441 7983000 Email: internet@uol.de Website: https://uol.de/en	University of Oldenburg Der Datenschutzbeauftragte Ammerländer Heerstr. 114-118 26129 Oldenburg Tel.: + 49 441 7984196 Email: dsuni@uol.de Website: https://uol.de/datenschutz/

To contact us, in particular to exercise your data subject rights, please contact the project coordinator named in the letterhead.

Joint controller	Data protection responsible
Prof. Dr. Stefan Debener Department of Psychology Lab. Neuropsychology Building A07-0-47 Tel.: 0441-798-4271 E-Mail: stefan.debener@uol.de	Carl von Ossietzky University of Oldenburg Data protection president/officer Ammerländer Heerstr. 114-118, 26129 Oldenburg Tel.: 0441-798-4196 E-Mail: dsuni@uol.de https://www.uni-oldenburg.de/datenschutz/

Legal basis

The legal basis for collecting your personal data is consent according to Art. 6.1 (a) of the General Data Protection Regulation (GDPR).

Data subject rights

- You have a right of access to your personal data (Art. 15 GDPR).
- You may without undue delay request the controller to correct any inaccurate personal data or to complete any incomplete personal data (Art. 16 GDPR).
- You have hereby been informed that you may request erasure of your personal data at any time (Art. 17 GDPR).

General information for participants

- You may request the restriction of processing if the legal requirements are met (Art. 18 GDPR).
- You have the right to receive your personal data in a structured, common and machine-readable format and to transfer this data to another controller (Art. 20 GDPR).
- You may object at any time to the processing of your personal data which is carried out on the basis of Article 6.1 (e) or (f) of the GDPR (Art. 21 GDPR).
- You have the right to withdraw your consent at any time with future effect, without affecting the lawfulness of processing based on consent before its withdrawal (Art. 7.3 GDPR).

Provision of data and consequences of non-provision

The provision of personal data relating to you is neither contractually nor legally required. You are not obliged to provide personal data concerning you. Failure to provide it would result in you not being able to participate in the study.

Right of complaint to a supervisory authority

If you believe that the processing of your personal data violates data protection regulations, please contact the data protection officer(s) of the responsible parties. Independently of this, you have a right to lodge a complaint with the competent supervisory authority.

The responsible supervisory authority is:

The State Commissioner for Data Protection of Lower Saxony

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